

Cost-Effective 50-km DAS Using a 200-kHz-Linewidth ECL with FDM Dual-Pulse Phase Noise Compensation

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Abstract We demonstrate a cost-effective 50-km DAS resilient to phase noise and fading. Leveraging a frequency-division-multiplexed dual-pulse delay-beat scheme, we achieve a strain sensitivity of $234 \text{ p}\epsilon/\sqrt{\text{Hz}}$ using a 200-kHz-linewidth ECL, and the performance using a 10-kHz-linewidth ECL is comparable to that of a 100-Hz-linewidth NLL. ©2026 The Author(s)

Introduction

Applying sensing function in the deployed optical fiber communication systems to provide added values such as fiber health monitoring and environmental sensing has attracted considerable interest. Among the sensing systems, Rayleigh backscattering (RBS)-based distributed acoustic sensing (DAS) using coherent detection has great potentials thanks to its high performance [1]. Accordingly, a number of integrated sensing and communication systems (ISAC) based on this scheme were proposed [2, 3]. However, a major obstacle to their widespread adoption is the requirement for high-cost components, particularly the narrow-linewidth lasers (NLLs), which are not standard in communication systems. Furthermore, a sensing range of at least 50 km is necessary to cover most existing optical communication systems [4]. Consequently, the development of long-haul DAS systems employing low-cost laser sources is highly desirable. The conventional approach to relaxing the laser requirement is phase noise compensation via an auxiliary interferometer that detects and recovers the phase noise [5, 6]. However, the performance is strongly dependent on the stability of the optical paths for the phase noise measurement, which is highly susceptible to mechanical vibrations in a practical environment, posing a significant challenge.

To address these issues, we propose a dual-pulse delay-beat scheme. As illustrated in Fig. 1 (a), the two probe pulses have different center frequencies, which enables the RBS signals to be separated at the receiver via appropriate digital filtering. One pulse is delayed relative to the other, with the time delay controlled directly by the modulator. The RBS trace is then split into two parts with a time delay between them, and

the two parts are simultaneously beaten with the local oscillator (LO). Spatial-domain phase differentiation is achieved from the phase difference between the two beat signals. As both signals contain the same laser phase noise introduced by the LO, such noise is effectively canceled in this process. We further optimize the scheme to extend the reach by employing the frequency division multiplexing (FDM) technique, as illustrated in Fig. 1 (b). Rather than transmitting two individual pulses, we transmit two groups of pulses to increase the pulse energy, where each group consists of multiple sub-pulses occupying distinct frequency bands. The proposed scheme also enables effective suppression of interference fading. In a proof-of-concept experiment, we demonstrated a DAS system over a 50 km fiber using an external cavity laser (ECL) with a 200 kHz linewidth, achieving a strain sensitivity of $234 \text{ p}\epsilon/\sqrt{\text{Hz}}$ at a spatial resolution of 20 m. Furthermore, the system performance was also evaluated using a 10-kHz-linewidth ECL, and the results show that its performance is comparable to that obtained with a 100-Hz linewidth NLL.

Operating principle

In DAS, spatial-domain phase differentiation is used to obtain vibration information within the tested region. In a dual-pulse scheme, there is a time delay between the pulses, corresponding to the distance delay of $2\Delta z$. The two traces generated by these two pulses are detected at the same time using a single receiver. At a given sampling time instant, the two traces contain the phase information of the fiber under test (FUT) which is separated by Δz and endure the same laser phase noise. By using the demodulated phase signal of one pulse as a reference, the

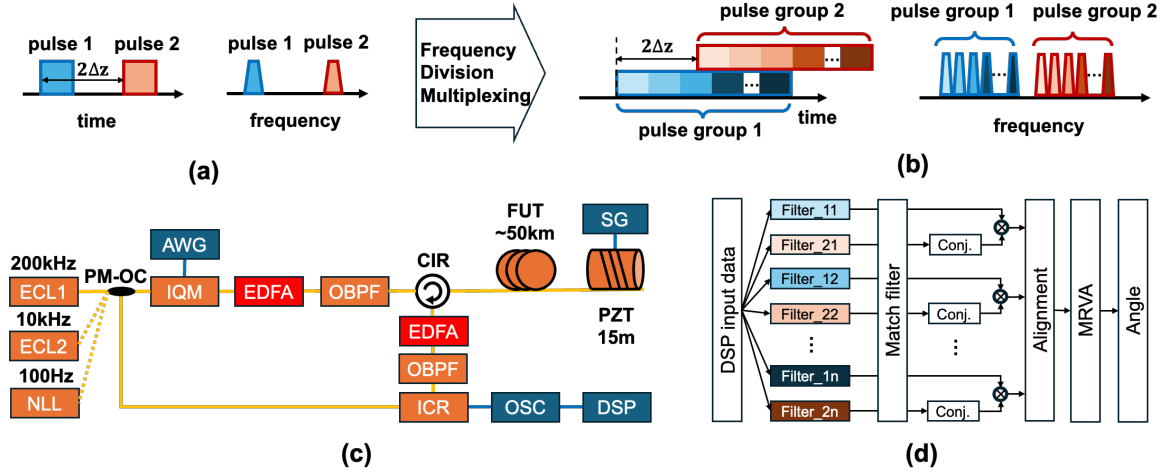


Fig. 1: Time and frequency domain of (a) dual-pulse scheme; (b) FDM dual-pulse scheme. (c) Experiment set-up. (d) Schematic diagrams of digital signal processing algorithm.

phase differentiation is achieved and the laser phase noise is removed by subtracting the phases of the two pulses signals [7]:

$$\Delta_z \varphi(t, z) = \Delta_z \varphi_R(t, z) - \Delta_z \theta(t, z) + \varphi_{21}(t) \quad (1)$$

where t and z are time and position respectively and Δ_z denotes spatial differentiation. The first term $\Delta_z \varphi_R(t, z)$ is the differential composite phase with the gauge length. It reflects the intra-pulse phase noise, which is negligible when narrow pulses are used. The second term $\Delta_z \theta(t, z)$ is the phase change caused by vibration within the region. The third term $\varphi_{21}(t) = \varphi_{0,2}(t) - \varphi_{0,1}(t)$ is related to the initial phase difference of the two pulses caused by linewidth and delay. It remains constant along distance and can be easily removed.

To increase pulse energy for an extended sensing range, we utilize the FDM technique. As illustrated in Fig. 1(b), two groups of pulses are transmitted, with a spatial delay of $2\Delta z$. Each group consists of multiple sub-pulses, where each sub-pulse occupies a distinct frequency band to avoid frequency crosstalk. Sub-pulses from the two groups with a spatial delay of $2\Delta z$ form individual pulse pairs for phase demodulation of the FUT. The measurement results from all pairs are first aligned and then averaged to suppress fading and improve the signal-to-noise ratio (SNR). The corresponding signal processing algorithm is depicted in Fig. 1(d).

Experiments

We conducted an experiment using the set-up shown in Fig. 1 (c). Here, we used an ECL (Santec TSL-570) centered at 1550.12 nm with a linewidth of around 200 kHz. For performance comparison, we also used an ECL (NLPM-A10-C34-10) with a linewidth of around 10 kHz and an NLL (NKT, E15) with a linewidth of around 100 Hz. The output was split into two parts by a

polarization-maintaining coupler (PM-OC), where 80% was used for the probe light generation and 20% was served as the LO. The arbitrary waveform generator (AWG) sampled at 1 GSa/s was used to generate FDM dual-pulse electrical signals and were sent to the IQ modulator (IQM). The pulse period was 600 μ s. An Erbium-doped fiber amplifier (EDFA) was used to boost the optical power, followed by an optical bandpass filter (OBPF) with a bandwidth of 0.8 nm to mitigate amplified spontaneous emission (ASE) noise of the EDFA. After passing through a circulator (CIR), the probe light was launched into a 50 km standard single-mode G.652D fiber with an attenuation coefficient of 0.2 dB/km. At the end of FUT, 15-m-long fiber was wrapped around a piezoelectric transducer (PZT) and was driven by a signal generator (SG) with 200 Hz sinewave signal to simulate the mechanical vibration. The RBS was amplified by another EDFA followed by an OBPF and were then detected using an integrated coherent receiver (ICR). The data was recorded by a 4-channel oscilloscope (OSC) at a sampling rate of 1 GSa/s and then processed offline.

The pulses were linear frequency modulated (LFM) with a chirp range from -200 to 200 MHz and a pulse duration of 5 μ s. The center frequencies of the two pulse groups were set to -250 and 250 MHz, respectively. The time delay between the two groups was 200 ns, corresponding to a gauge length of 20 m. The spectrum of the RBS is depicted in Fig. 2 (a). Following the algorithm shown in Fig. 1 (c), each pulse group can be decomposed into multiple sub-pulses, forming multiple pulse pairs. For a given pulse duration and chirp range, a larger number of sub-pulses leads to a smaller chirp bandwidth, lower spatial resolution, and lower energy or SNR of individual sub-pulses, yet yields superior fading suppression and reduced impact of intra-pulse phase noise.

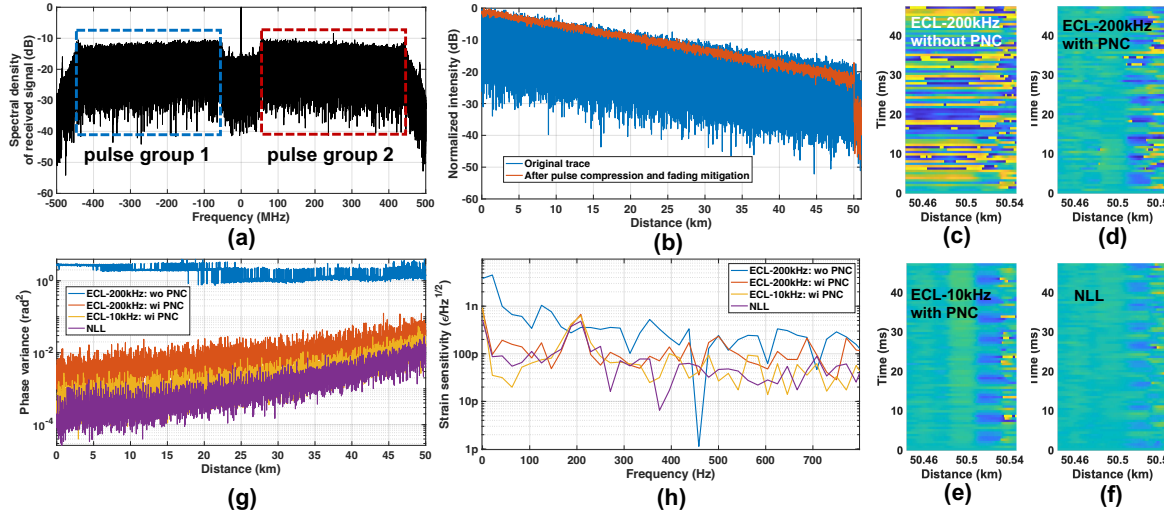


Fig. 2: (a) RBS spectrum. (b) Normalized signal intensity before and after pulse compression and fading mitigation. Waterfall plot of measurement result at the vibration region using (c) 200-kHz-linewidth ECL without PNC, (d) 200-kHz-linewidth ECL with PNC, (e) 10-kHz-linewidth ECL with PNC, and (f) 100-Hz-linewidth NLL without PNC. (g) Variance of demodulated phase along distance. (h) Strain sensitivity at the vibration region.

The optimized number of pulse pairs was found to be 20, yielding a sub-pulse duration of 250 ns. After digital filtering, pulse compression was achieved via matched filtering. The complex conjugate of the sub-pulse from one group was calculated and multiplied by that from the other group to extract the phase difference. Following time alignment, the moving rotated-vector average (MRVA) algorithm [8] with a 200 ns sliding window was employed to suppress interference fading. The signal intensity before and after pulse compression and fading mitigation is presented in Fig. 2(b), where a clear improvement in SNR can be observed. Finally, the phase of the FUT was demodulated. For performance evaluation, we compared four cases using:

- ECL with 200 kHz linewidth without phase noise compensation (PNC). Results were calculated using only pulse group 1. (Plotted in blue).
- ECL with 200 kHz linewidth with PNC. (Plotted in red).
- ECL with 10 kHz linewidth with PNC. (Plotted in yellow).
- NLL with 100 Hz linewidth without PNC. (Plotted in purple).

The variance of the demodulated phase along distance is depicted in Fig. 2(g). It can be observed that, with a 200-kHz-linewidth ECL, the system performance is improved by approximately two orders of magnitude when PNC is applied. A performance gap still exists compared with the case using a 100-Hz-linewidth NLL. Nevertheless, when the laser linewidth is further reduced to 10 kHz, the resulting performance becomes almost comparable to that achieved with a 100-Hz-linewidth NLL.

The strain sensitivity at the vibration region is

presented in Fig. 2 (h). The corresponding waterfall plots for each case are shown in Fig. 2 (c)–(f). A severe noise is observed for the 200-kHz-linewidth ECL case without PNC, whereas a clear vibration signal is recovered after applying PNC, yielding a strain sensitivity of approximately $234 \text{ p}\epsilon/\sqrt{\text{Hz}}$. For the 10-kHz-linewidth ECL and the 100-Hz-linewidth NLL case, cleaner vibration signals are obtained, with strain sensitivities of approximately $100 \text{ p}\epsilon/\sqrt{\text{Hz}}$ in both cases.

Conclusions

We experimentally demonstrate a frequency-division-multiplexed dual-pulse delay-beat coherent DAS system, which achieves a measurement range of 50 km and a strain sensitivity of $234 \text{ p}\epsilon/\sqrt{\text{Hz}}$, using a low-cost ECL with 200kHz linewidth. With laser phase noise compensation (PNC), the system performance is enhanced by approximately two orders of magnitude. Moreover, with PNC, the performance obtained using a 10-kHz-linewidth ECL is nearly comparable to that of a 100-Hz-linewidth NLL. These results indicate that the cost of a coherent DAS system can be substantially reduced by adopting low-cost laser sources without obvious degradation in sensing performance, offering a cost-effective solution for high-performance long-haul DAS in practical field applications.

Acknowledgements

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